

CSE5026: Foundation and Frontiers of Cognitive Science

Perception and Attention

Yang Xu 徐炆

Department of Computer Science and Engineering
Southern University of Science and Technology

Overview

- Perception 感知
- Perceptual Constancy 感知 (知觉) 恒常性
- Organization of Perception 感知的组织
- Attention 注意

Perception 感知/知觉

- Perception (Latin, *perception*, “gathering, receiving”) is the organization, identification, and interpretation of sensory information in order to represent and understand the presented information or environment.
(from *Psychology*, Schacter D, 2011)
- 感知/知觉：**外界**刺激作用于**感觉器官**时，**脑**对外界的整体的看法和理解，
为我们对外界的感官信息进行组织和解释
- 可看作一组过程，获取、理解、筛选、组织信息
- 《说文解字》 “感：动人心也”



“感” 字上半部分
“咸” 的甲骨文，象
武器劈砍状

Perception system classified (by sensory organs 感官)

名称	注意的方式	感受单位	器官的解剖学部位	器官的活动性
基本方位系统	一般定向	机械感受器	前庭器官	身体平衡
听觉系统	倾听	机械感受器	包括中耳和外耳在内的耳蜗器官	声音方位的确定
触觉系统	接触	机械感受器及可能的温度感受器	皮肤（及其附属物或开口处）、关节（包括韧带）、肌肉（包括肌腱）	多种类的探索
嗅觉系统	嗅	化学感受器	鼻腔（鼻）	闻气味
味觉系统	尝味	化学和机械的感受器	口腔（口）	尝滋味
视觉系统	看	光感受器	视觉机构（眼睛及其内部的和外部的眼肌、如与前庭器官、头部和整个身体相连的肌肉）	适应瞳孔的调节、视网膜感光变化

Perceptual Systems by Sensory Organ

System	Mode of attention	Receptors	Anatomical structures	Activity
Basic orienting	General orientation	Mechanoreceptors	Vestibular organs	Body equilibrium
Auditory	Listening	Mechanoreceptors	Cochlear organs, with the middle and outer ear	Sound localization
Haptic	Touching	Mechanoreceptors and possibly thermoreceptors	Skin (including appendages and openings), joints (including ligaments), muscles (including tendons)	Active exploration
Olfactory	Smelling	Chemoreceptors	Nasal cavity (nose)	Sniffing odors
Gustatory	Tasting	Chemical and mechanical receptors	Oral cavity (mouth)	Tasting flavors
Visual	Looking	Photoreceptors	Eyes and internal and external eye muscles, linked with the vestibular organs, head, and body	Pupil adjustment and changes in retinal stimulation

Composition of Perception

- Sensation $\Rightarrow$ **Perception** $\Rightarrow$ Representation
- **Sensation (low level)**: When an object stimulates a sense organ, the brain registers individual properties of that object.
- **Perception (mid level)**: The integration of sensations into a more complete and complex experience.
- **Representation (high level)**: A mental image or impression formed through repeated perception of the same object or class of objects.

Sensory Thresholds

- Not every stimulus produces a sensation
- A stimulus must reach sufficient intensity to be detected.
- Absolute threshold: The minimum stimulus intensity a person can detect.
- Psychophysics studies how changes in physical stimuli relate to changes in sensation across vision, hearing, taste, touch, and smell.

Weber's Law



- E. H. Weber (1795–1878), German physiologist.
- Absolute threshold: The minimum intensity needed to detect a stimulus.
- **Difference threshold:** The smallest detectable difference between two stimuli, also called the just noticeable difference (JND).
- Weber's law: Within a certain range, the difference threshold is approximately a constant fraction of the initial stimulus intensity.

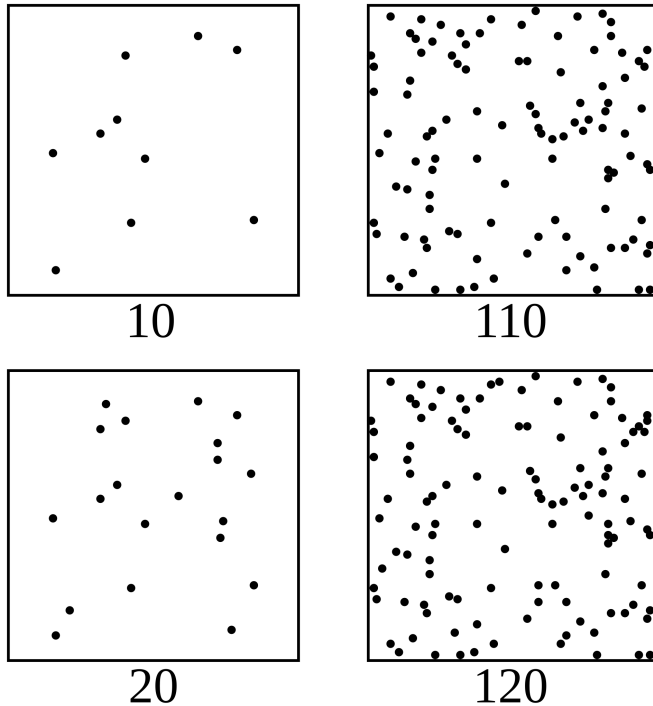
$$\frac{\Delta I}{I} = k$$

ΔI : difference threshold

I : initial intensity

k : Weber fraction (constant)

Weber's Law 韦伯定律



The lower square contains 10 more dots than the upper one. However, the perception is different.

$$\Delta I = 20 - 10 = 10 = 120 - 110$$

$$I_1 = 10 \Rightarrow k_1 = \frac{\Delta I}{I_1} = \frac{10}{10} = 1$$

$$I_2 = 110 \Rightarrow k_2 = \frac{\Delta I}{I_2} = \frac{10}{110} \approx 0.01$$

Source: https://en.wikipedia.org/wiki/Weber-Fechner_law

Weber's Law

- Typical Weber fractions
- Initial stimulus intensities appear in parentheses k

$$\frac{\Delta I}{I} = k$$

ΔI : difference threshold
 I : initial intensity
 k : Weber fraction (constant)

Stimulus (initial intensity)	Weber fraction
Pitch (2000 Hz)	1/333
Deep pressure (400 g weight)	1/77
Visual brightness (1000 photons)	1/62
Lifted weight (300 g)	1/53
Loudness (100 dB, 1000 Hz)	1/11
Rubber odor (200 odor units)	1/10
Skin pressure (5 g/mm ²)	1/7
Saltiness (3 mol/kg)	1/5



Weber-Fechner Law

- A modification to the original Weber's Law by German psychologist Gustav Fechner (1801-1887)
- Perceived intensity increases *logarithmically* with stimulus intensity. As the physical stimulus intensity increases, the perceived intensity grows slower.

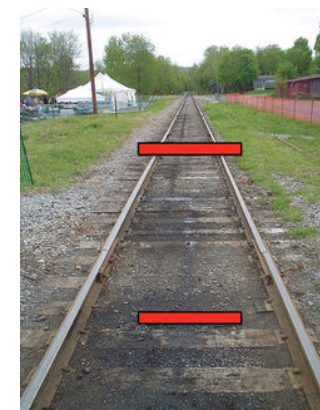
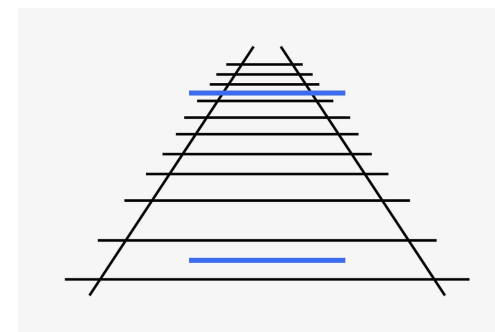
$$p = \alpha \ln \frac{S}{S_0} \quad p: \text{Perception. } S: \text{Stimulus intensity. } S_0: \text{initial intensity.}$$

Overview

- Perception 感知
- **Perceptual Constancy 感知 (知觉) 恒常性**
- Organization of Perception 感知的组织
- Attention 注意

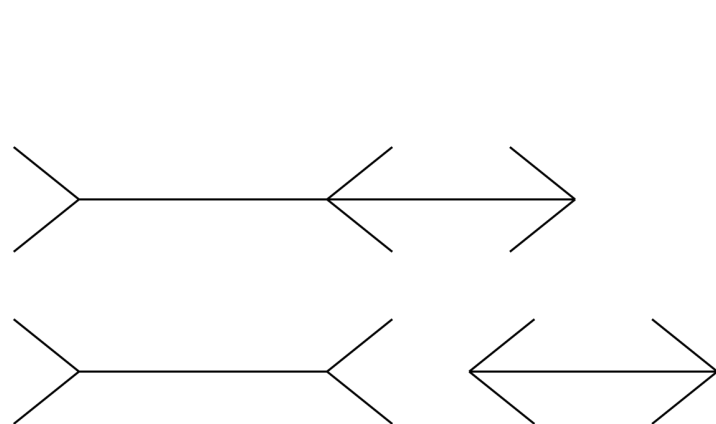
Perceptual Constancy

- Perceptual constancy: Object properties, such as size, shape, and color, appear relatively stable despite changes in viewing conditions and sensory input.
- Size constancy:
 - Perceived size remains relatively stable as retinal image size changes.
 - A person at 3, 5, 15, or 30m distance away can appear similar in size.
- Ponzo illusion 庞佐幻觉
 - Converging lines provide depth cues.
 - At different depths, equal *retinal sizes* (视网膜上的尺寸) can imply different object sizes.

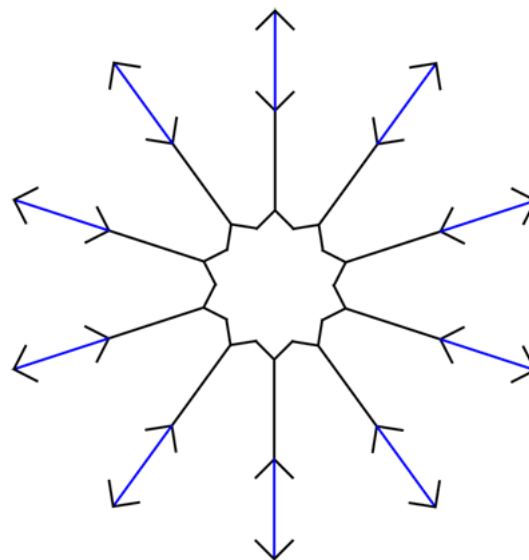


Perceptual Constancy 感知恒常性

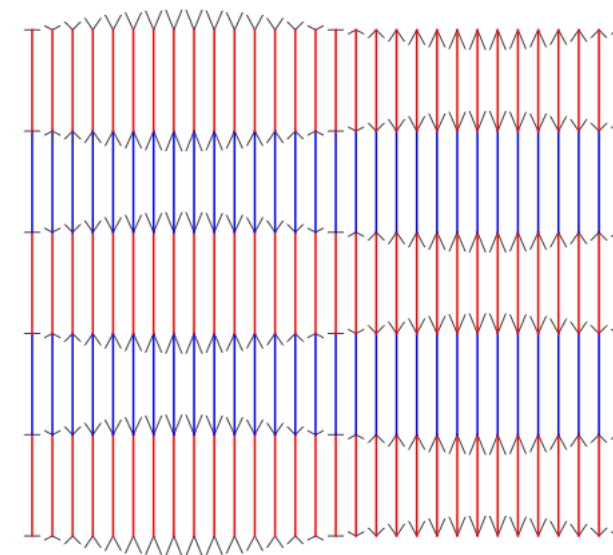
- Müller-Lyer illusion 缪勒-莱尔错觉



Sarcone's Dynamic Müller-Lyer Illusion



Sarcone's Dynamic Müller-Lyer Illusion
involving sinusoidal wave motion with dynamic color spreading effects



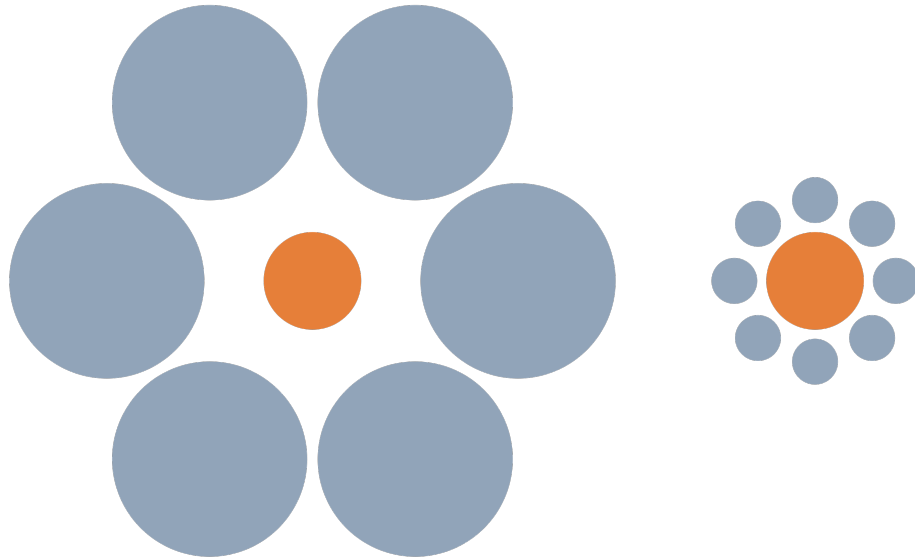
Source: https://en.wikipedia.org/wiki/Müller-Lyer_illusion



Hermann Ebbinghaus,
1850-1959, German
psychologist

Perceptual Constancy

- Relative size illusion (Ebbinghaus illusion or Titchener circles)

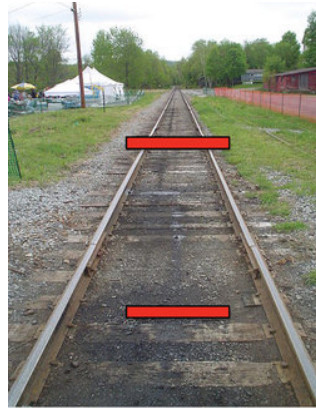
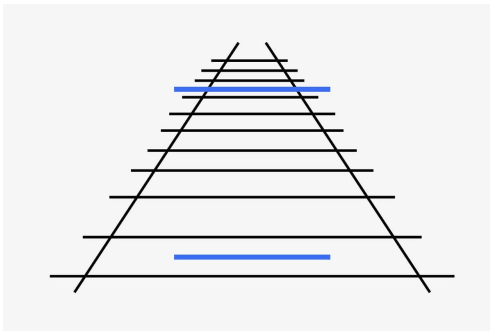


The two central circles are the same size. The surrounding circles change their apparent size.

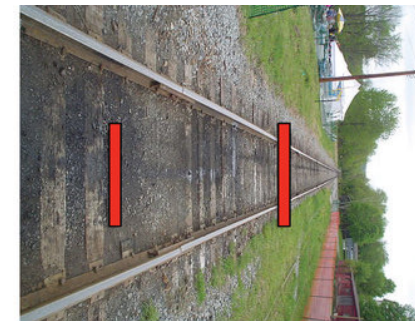
Source: <https://zh.wikipedia.org/zh-cn/艾賓浩斯錯覺>

Factors Affecting Size Constancy

- Context: Richer depth cues generally support stronger size constancy.

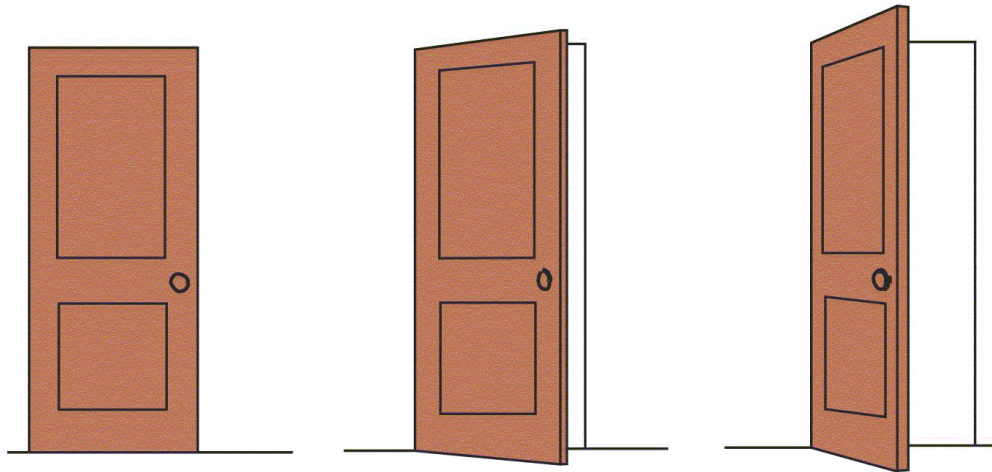


- Constancy can weaken at long distances.
- Horizontal viewing often strengthens constancy.



Shape Constancy

- An object's shape appears relatively stable as the viewing angle or object orientation changes, even though its retinal image changes.



Color Constancy (颜色恒常性)

- An object's perceived color remains relatively stable under different lighting conditions.
- A red flag can still look red under yellowish or bluish light.
- A green apple can look green at midday and at sunset.



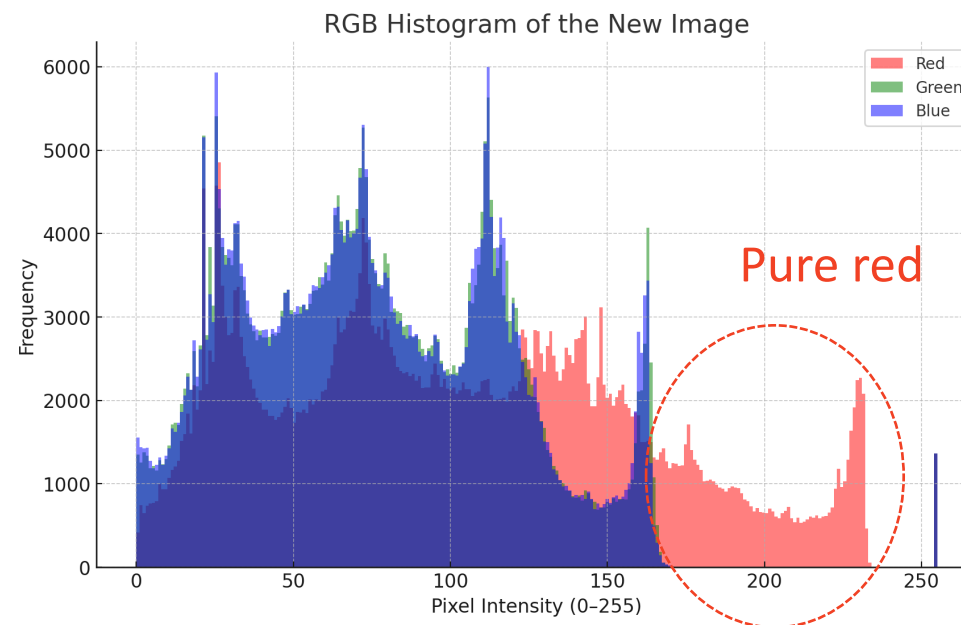
Color constancy makes this image appear to have **red**, **green** and **blue** hues (especially in dark),

even though it is composed of only light and dark shades of red and white.

Source: https://en.wikipedia.org/wiki/Color_constancy

Color Constancy

- Pixel RGB values



Highly overlapping G and B distribution

- Pure *white*: (255, 255, 255)
- Light *gray*: (230, 230, 230)
- Medium *gray*: (180, 180, 180)
- Dark *gray*: (100, 100, 100)

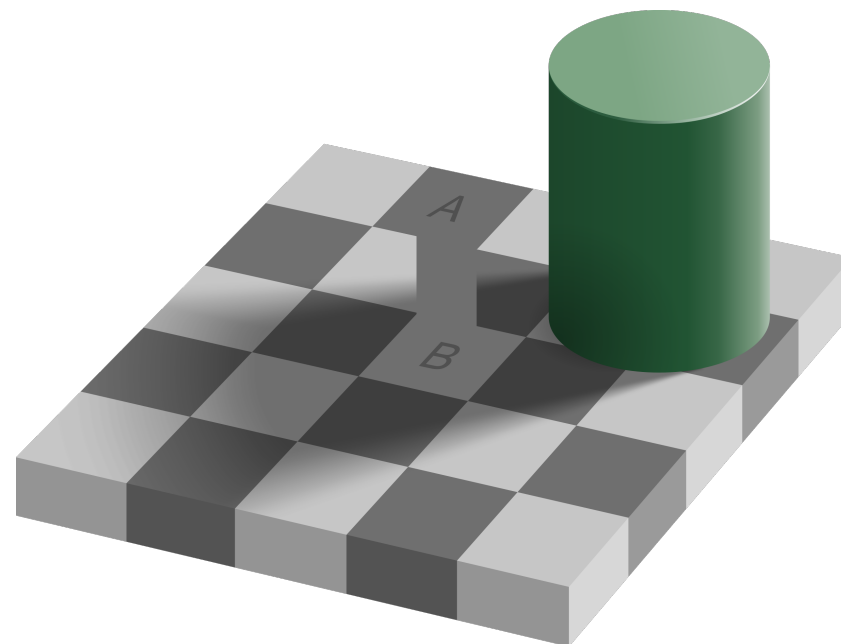
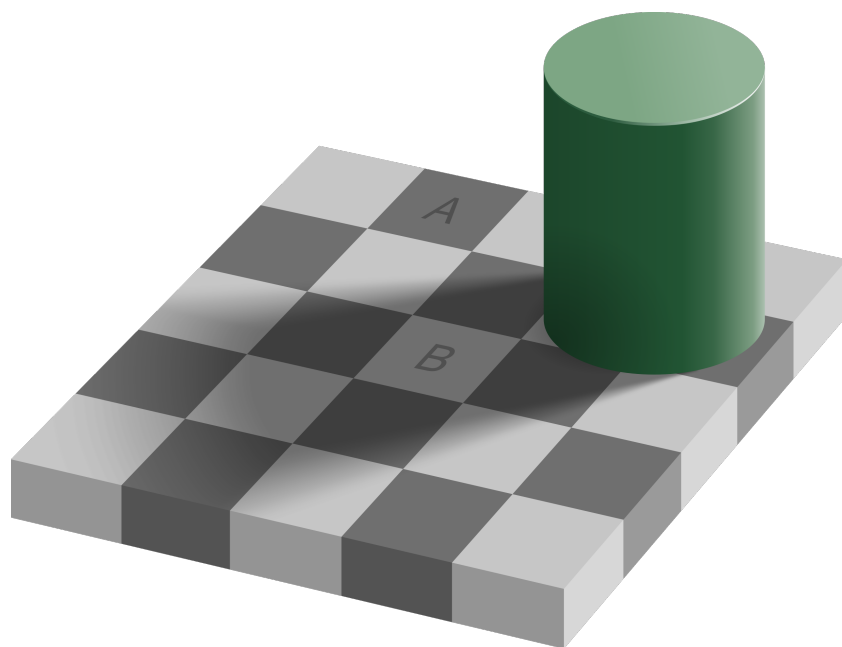
Color Constancy

- Checker shadow illusion

Source:

https://en.wikipedia.org/wiki/Color_constancy

https://en.wikipedia.org/wiki/Checker_shadow_illusion



Constancy makes A appear darker than B.

Perceptual constancy in VLM?

- Vision-Language Models (VLMs)'s capability of perceptual constancy varies.
- Overall: VLMs lag behind human. VLMs perform good in **shape** constancy, but not in **color** or **size**.



Is the actual shape of the tabletop in the picture a parallelogram? Please answer with Yes or No and provide an explanation.

Correct Answer: No

GPT4o: No. The actual shape of the tabletop in the picture is not a parallelogram. It is a rectangle. The tabletop in the image has four right-angle corners, characteristic of a rectangle, rather than having slanted edges or angles. This confirms the shape as a rectangle and not a parallelogram.

Source: Sun et al. (2025) <https://arxiv.org/pdf/2502.10273v1>

Perceptual constancy in VLM?

- Size



Is the actual size of the missile in the front larger than the size of the missile in the back? Please answer with Yes or No and provide an explanation.

Correct Answer: No

GPT4o: NO. the actual size of the missile in the front is not larger than the size of the missile in the back. The two missiles shown in the image are identical in size they appear to be the same model (df-5b) loaded on similar transport vehicles.



In the picture, is the actual width of the pier consistent from near to far? Please answer with Yes or No and provide an explanation.

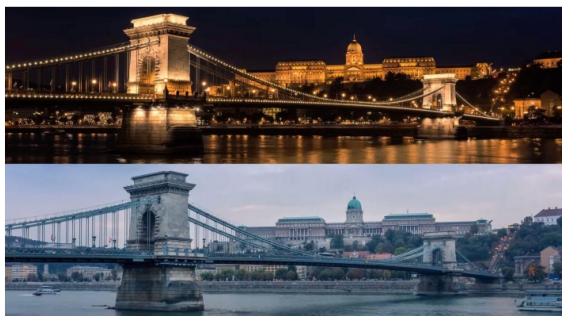
Correct Answer: Yes

GPT4o: No. The actual width of the pier remains consistent from near to far. The apparent decrease in width is due to perspective. As objects extend into the distance, they appear smaller and narrower, creating an illusion that they are converging.

Source: Sun et al. (2025) <https://arxiv.org/pdf/2502.10273v1>

Perceptual constancy in VLM?

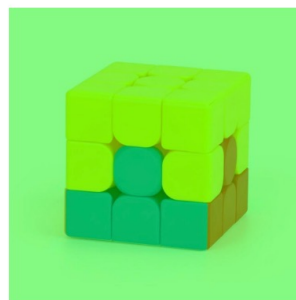
- Color



In the picture, has the actual color of the bridge itself changed? Please answer with Yes or No and provide an explanation.

Correct Answer: No

GPT4o: No. The actual color of the bridge itself has not changed. The difference in appearance between the two photos is primarily due to the lighting conditions. The top image is taken at night with artificial lighting illuminating the bridge and surrounding area. The bottom image is taken during the day with natural daylight without the effects of artificial lighting.



What color is the middle square of the Rubik's Cube? Please select from the following choices:

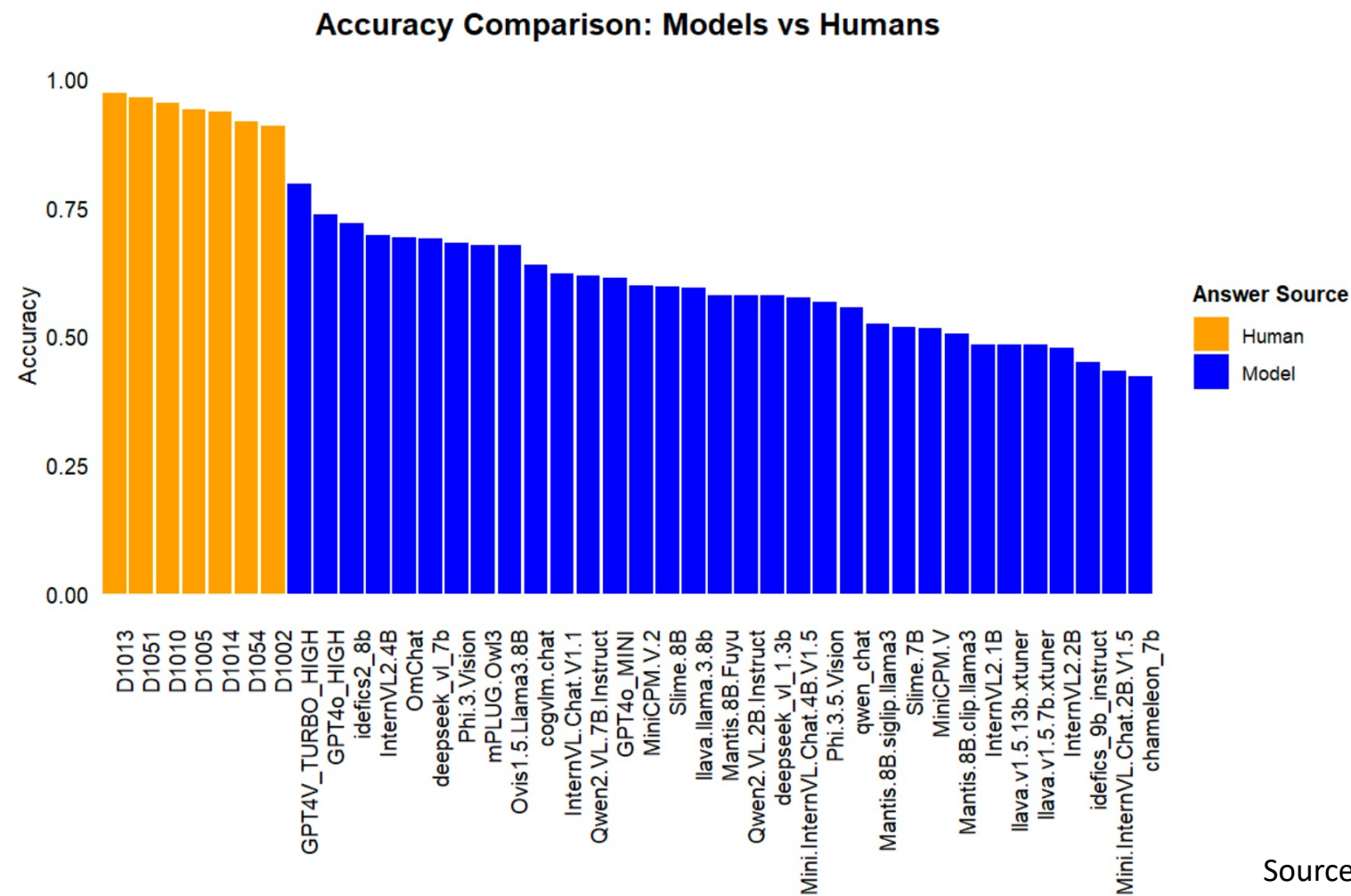
A: Green,
B: Yellow,
C: Blue

Correct Answer: C

GPT4o: A green. The middle square of the rubik's cube shown in the image is color green this can be determined by observing the central square on the visible face of the cube

Source: Sun et al. (2025) <https://arxiv.org/pdf/2502.10273v1>

Perceptual constancy in VLM?



Source: Sun et al. (2025) <https://arxiv.org/pdf/2502.10273v1>

Overview

- Perception 感知
- Perceptual Constancy 感知 (知觉) 恒常性
- **Organization of Perception 感知的组织**
- Attention 注意

Organization of Perception

- Gestalt psychology (early 1900s)
- Emphasizes the organization of experience and behavior as wholes. Challenges structuralism's analysis into elements and behaviorism's stimulus–response account.
- A whole is more than a collection of parts. Consciousness is more than a set of sensations, and behavior is more than a chain of reflexes.

$$\text{Whole} \neq \sum \text{parts} \quad \text{consciousness} \neq \{\text{senses}\}$$

- Gestalt psychology began with studies of perception, but extends beyond perception.

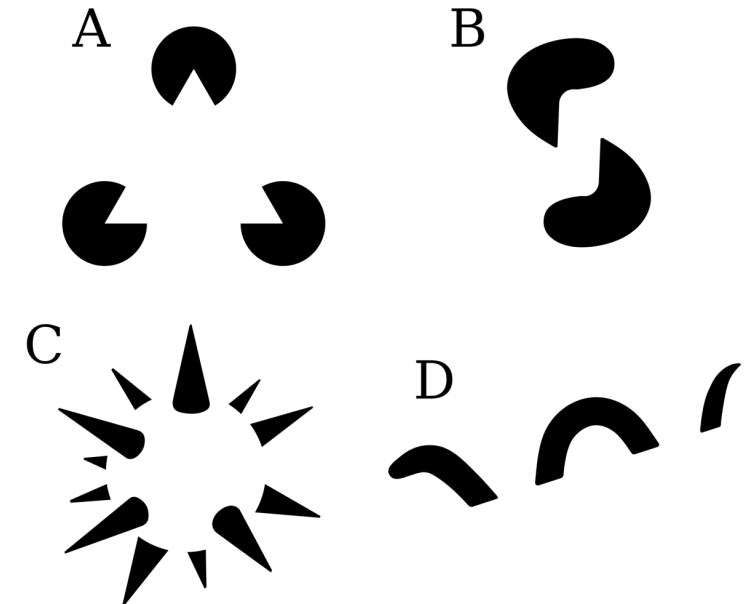
Gestalt Theory

Source: https://en.wikipedia.org/wiki/Gestalt_psychology_-_Principles

- Gestalt is a German word for an *organized form or configuration*, often described as a dynamic whole.
- Max Wertheimer (1880–1943)
Wolfgang Köhler (1887–1967)
Kurt Koffka (1886–1941)
- **Holism**: The brain organizes experience into **wholes (整体)**
 - *Holism*: “The whole is something else than the sum of its parts, because summing is a meaningless procedure, whereas the whole-part relationship is meaningful.”

Principles of Gestalt Systems

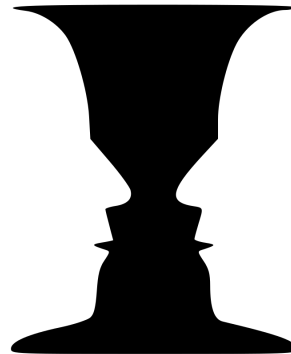
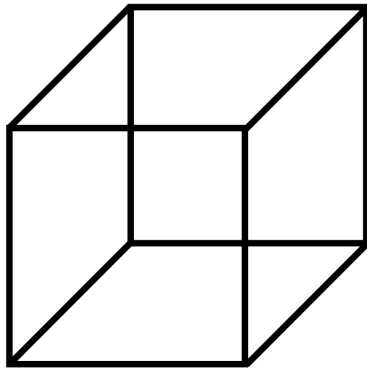
- **Reification** (具体化, 实体化)
- Perception is constructive and generative.
- Perception adds spatial structure beyond the sensory input.
 - A: An apparent triangle
 - C: An apparent three-dimensional sphere
 - B: Columns.
 - D: A ground surface
 - These perceived structures are not explicitly drawn.



Source: https://en.wikipedia.org/wiki/Gestalt_psychology_-_Principles

Principles of Gestalt Systems

- **Multistability** (多稳态)
- An ambiguous stimulus can support two or more interpretations. Perception may switch repeatedly between them.



Source: https://en.wikipedia.org/wiki/Gestalt_psychology_-_Principles

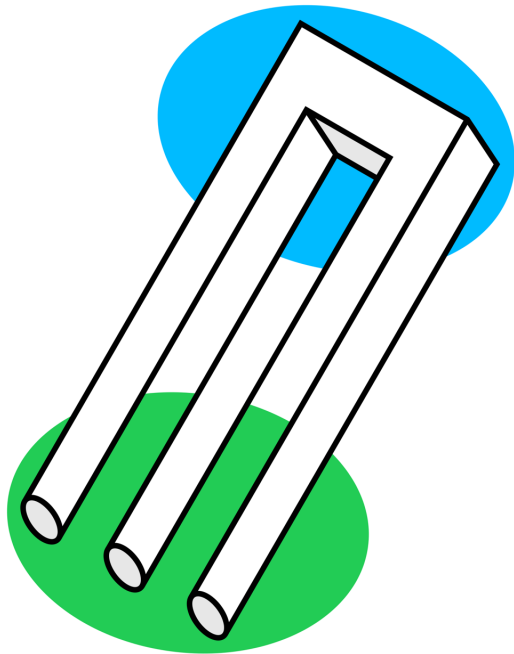
Principles of Gestalt Systems

Source:

https://en.wikipedia.org/wiki/Necker_cube

https://en.wikipedia.org/wiki/Impossible_trident

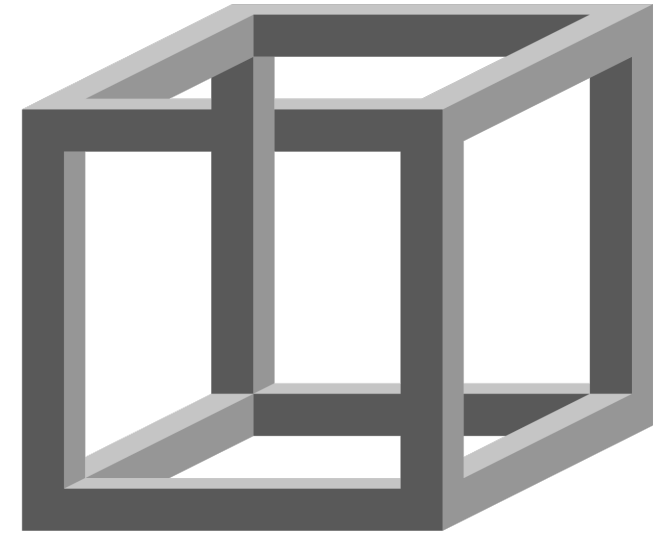
- More examples on multistability



Impossible trident



Undecidable monument

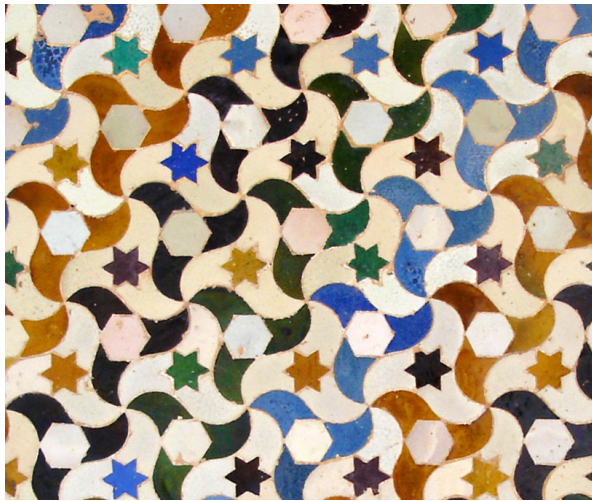


Impossible cube

Principles of Gestalt Systems

多稳态

- More examples on multistability (between background and foreground)



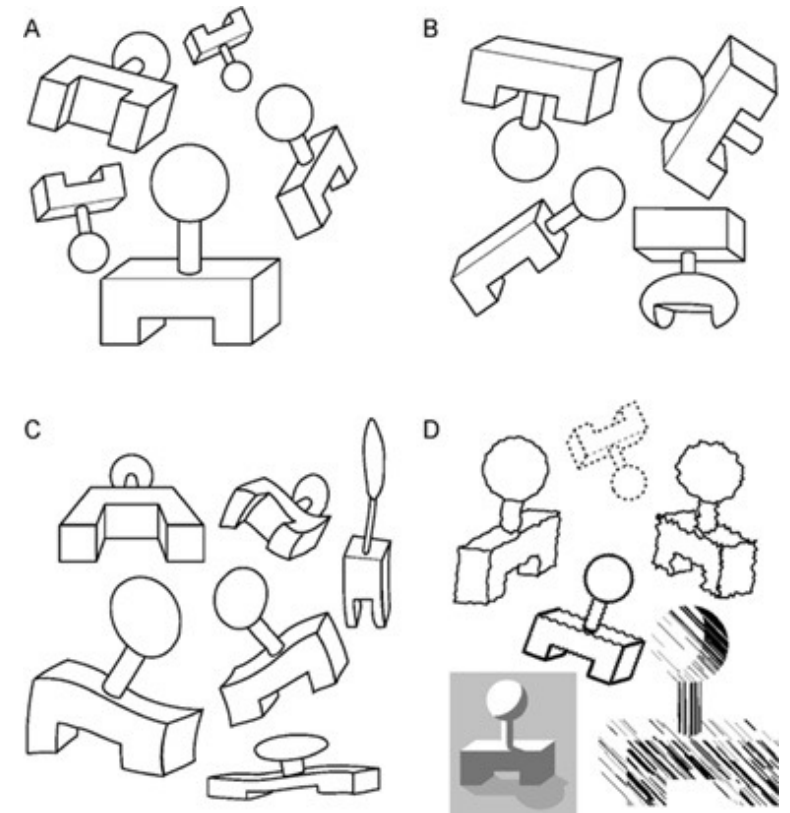
Math term: exact insertion
Gestalt term: multistability
of perception

M. C. Escher's art, inspired by tessellation

Source: https://en.wikipedia.org/wiki/M._C._Escher

Principles of Gestalt Systems

- **Invariance** (不变)
- We can recognize geometric forms across rotation, translation, scaling, deformation, changes in lighting, and changes in their visual elements.
 - A: The forms are clearly distinct from those in B.
 - C: Rotation and deformation preserve recognition.
 - D: Different drawing elements still convey the same forms.



Source: https://en.wikipedia.org/wiki/Gestalt_psychology_-_Principles

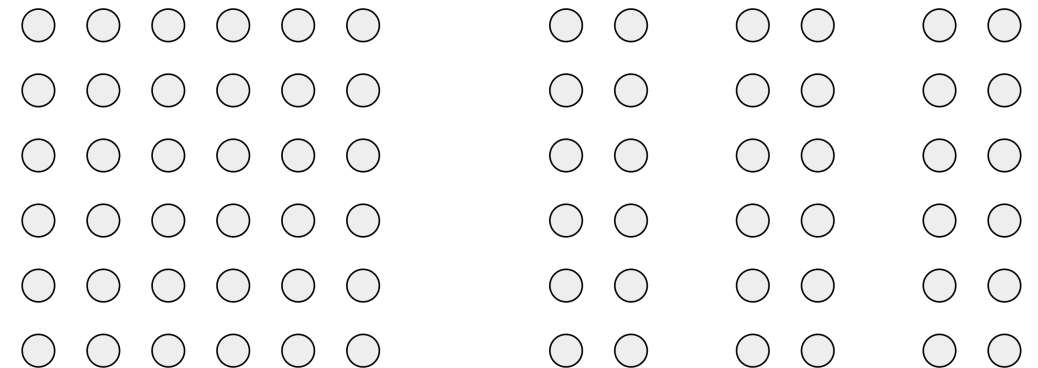
Perceptual Organization: Law of Prägnanz (简洁律)

- Gestalt psychologists pioneered the study of perceptual organization: How perception is organized.
- The law of Prägnanz is a central principle of perceptual organization.
 - *Prägnanz* is a German word that directly translates to 'pithiness' and implies salience, conciseness, and orderliness
- We tend to perceive forms as regular, orderly, symmetrical, and simple.
- Related grouping principles include:
Proximity, similarity, closure, symmetry, and continuity.
(邻近律) (相似律) (封闭律) (对称律) (连续律)

Source: https://en.wikipedia.org/wiki/Gestalt_psychology_-_Principles

Law of Prägnanz

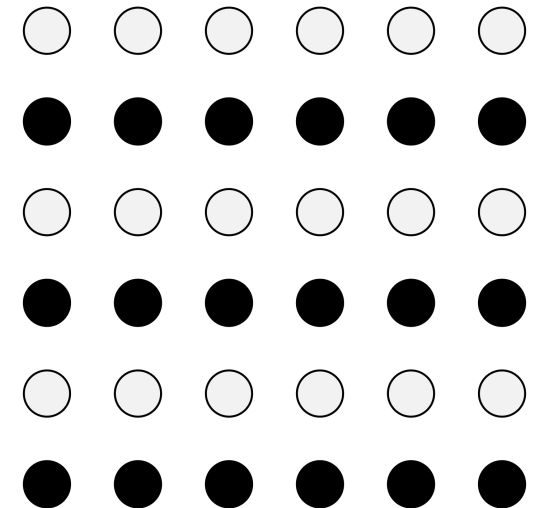
- Law of proximity (邻近律)
- We tend to group objects that are close together.
- The image contains 72 circles, but we perceive groups:
 - One group of 36 on the left
 - Three groups of 12 on the right



Source: https://en.wikipedia.org/wiki/Gestalt_psychology_-_Principles

Law of Prägnanz

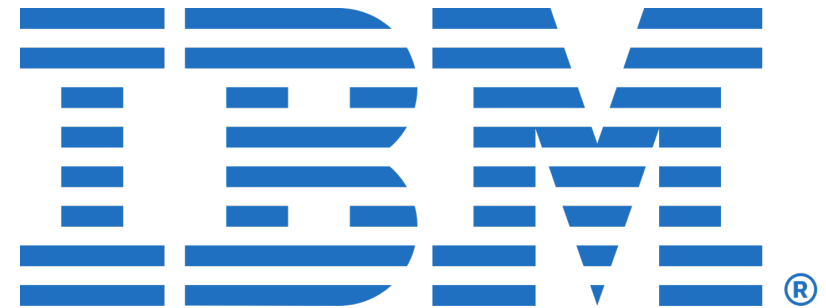
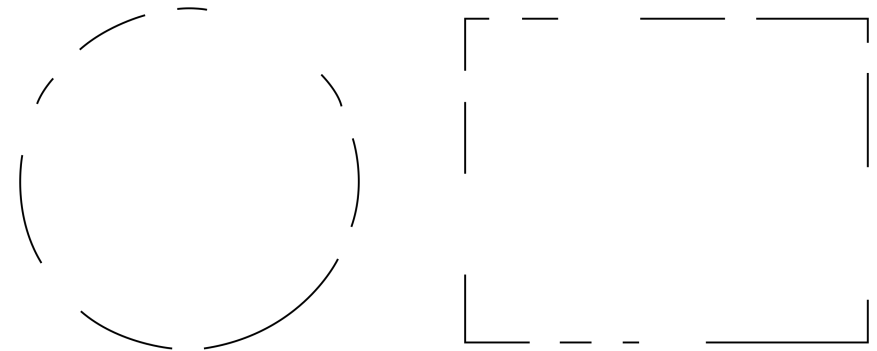
- Law of similarity (相似律)
- We group objects that share features such as shape, color, or size.
- The 36 evenly spaced circles include 18 dark and 18 light circles.
- Grouping by shade produces six apparent horizontal rows.



Source: https://en.wikipedia.org/wiki/Gestalt_psychology_-_Principles

Law of Prägnanz

- Law of closure (封闭律)
- We tend to perceive complete objects despite gaps in their contours.
- Incomplete shapes, letters, and pictures can appear whole.



Source: https://en.wikipedia.org/wiki/Gestalt_psychology_-_Principles

Law of Prägnanz

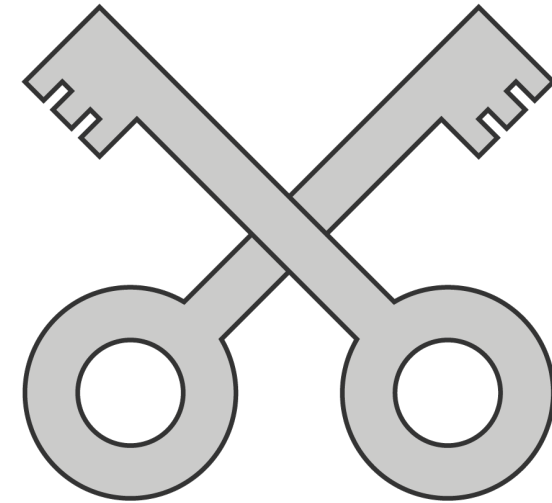
- Law of symmetry (对称律)
- We organize elements into symmetrical forms around a center. Symmetrical divisions can be perceptually pleasing.
- Separate symmetrical elements can appear to belong to one coherent shape.
- We tend to see three pairs of brackets rather than six individual brackets.

() [] ()

Source: https://en.wikipedia.org/wiki/Gestalt_psychology_-_Principles

Law of Prägnanz

- Law of continuity (连续律)
- When objects cross, we tend to perceive each as a continuous whole.
- Smooth continuation favors grouping. We see the partly hidden key as one continuous key rather than two separate halves.



Source: https://en.wikipedia.org/wiki/Gestalt_psychology_-_Principles

Chinese Characters and Gestalt Psychology

中国文字与格式塔心理学

Gestalt: The whole is different from the sum of its parts.

六书之一的“会意”字：由两个或两个以上的表形或表义部件组合在一起，表示新的意义。

Associative compounds (会意), one of the six traditional categories, combine two or more pictorial or semantic components to express a new meaning.

𡥉 (好), “女”面对“子”, 表示相亲相爱

好: woman + child
An image of affection

𠂔 (兵), 由两个𠂔 (又) 和一个𠂔 (斤) 组成

兵: two hands + an axe
An image of a weapon

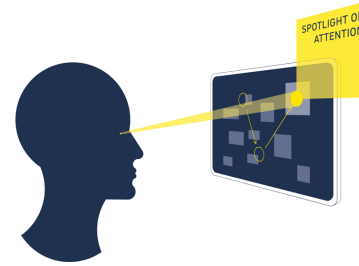
又如“信”, 由“人”和“言”组成, 表示诚信

信: person + speech
An image of trustworthiness

Overview

- Perception 感知
- Perceptual Constancy 感知 (知觉) 恒常性
- Organization of Perception 感知的组织
- **Attention 注意**

Attention



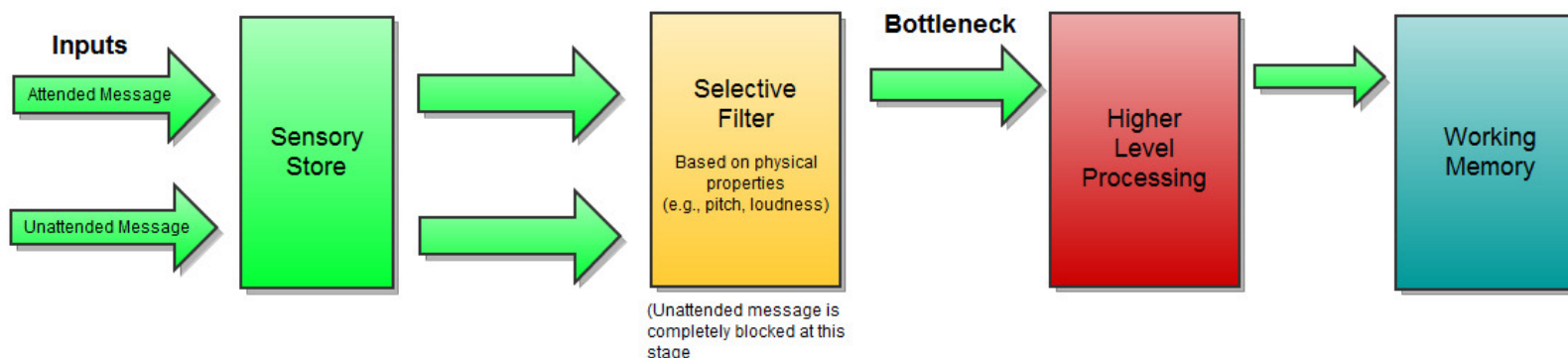
- **Attention: Directing awareness**
- Selective focus on a particular object or event.
 - In Psychology, traditionally treated as a state supporting other mental processes, without separately representing objects or their properties.
- In Cognitive Science:
 - Attention is an internal mechanism that selectively controls and regulates stimulus processing
 - **Multiple models (theories) of selective attention**

Models of Attention

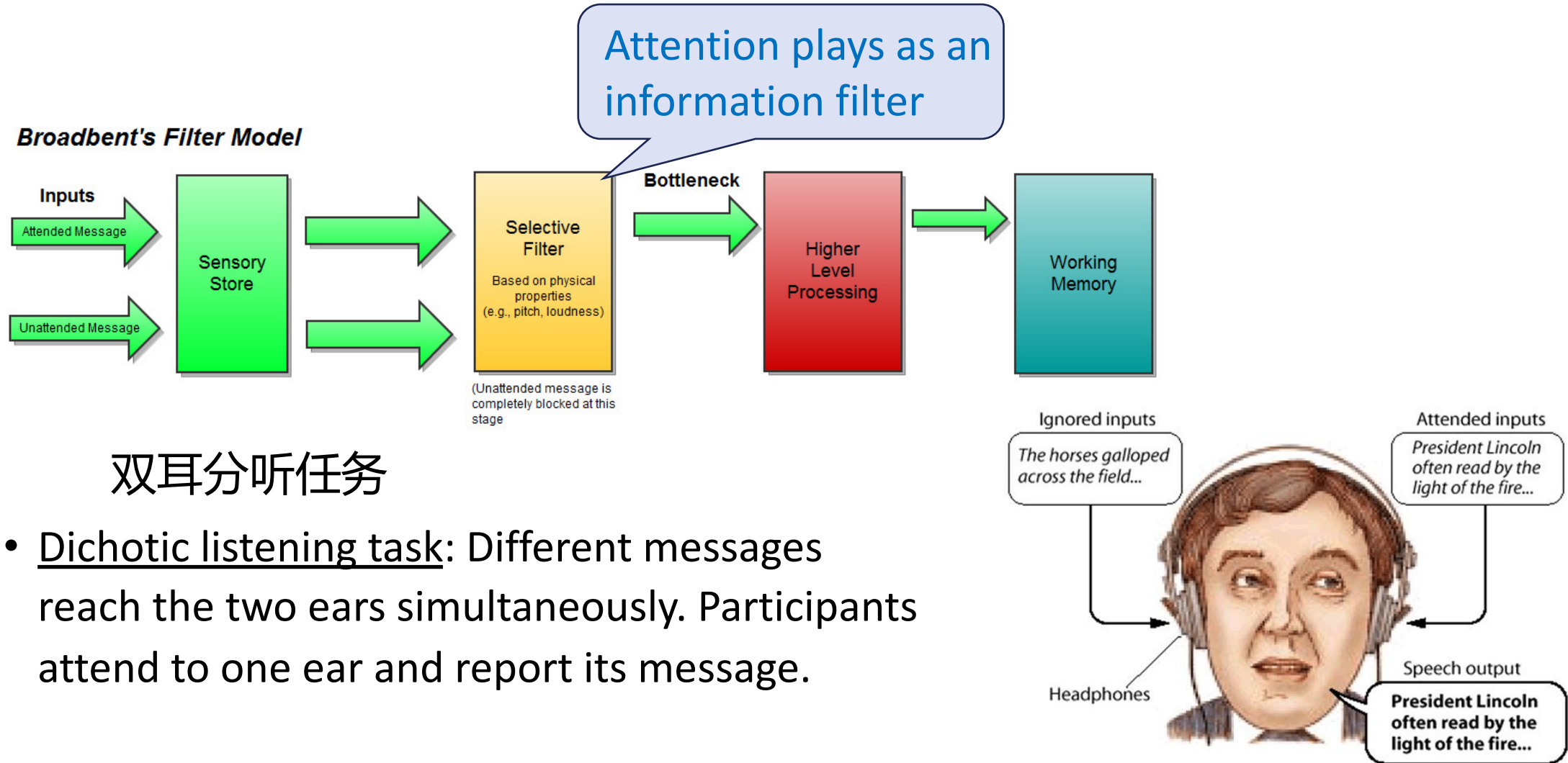
- **Broadbent's filter model** (D. Broadbent, British psychologist)
- A filter prevents overload by selecting information for further analysis. Other information briefly remains in sensory memory and then decays.

感知记忆

Broadbent's Filter Model

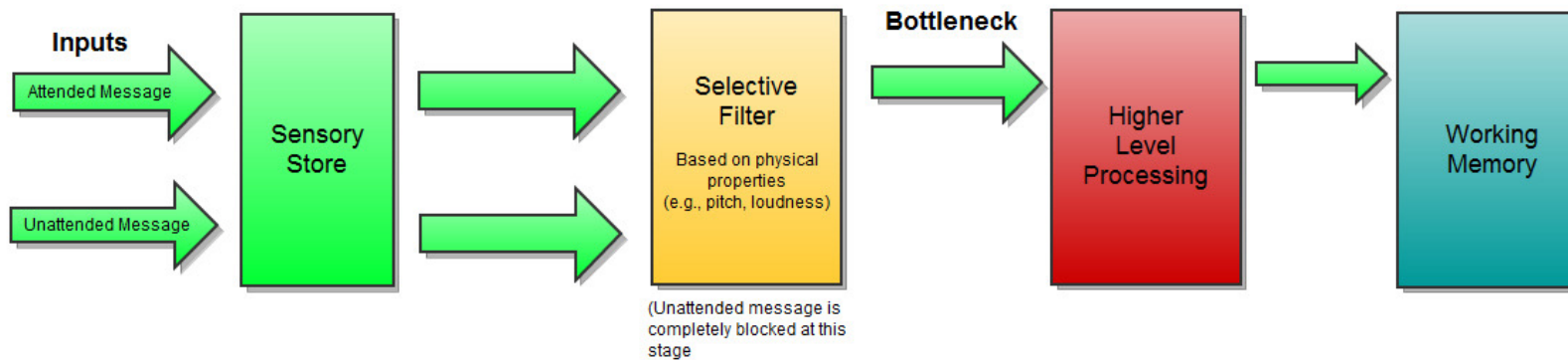


Filter Model of Attention



Filter Model of Attention

Broadbent's Filter Model



- An all-or-none filter admits one channel at a time.
- **Early selection:** Physical features (low level) guide selection before analysis of meaning.
- Weakness: The cocktail party effect challenges a strict early filter.

Cocktail Party Effect

- We can follow a friend's conversation in a noisy room.
- Our own name in another conversation may capture attention.
- Speech in our native language may stand out among conversations in unfamiliar languages.



Source: https://en.wikipedia.org/wiki/Cocktail_party_effect

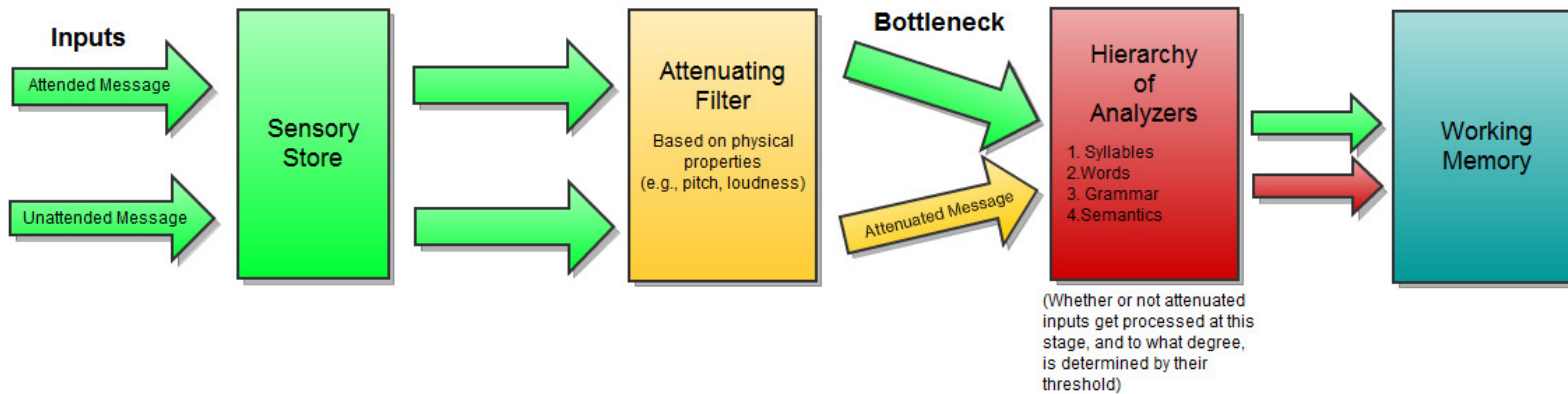
Attenuation Model 衰减模型 An improvement of filter model

(British psychologist)

- Anne Treisman (1964, 1969) challenged an all-or-none filter.
- In a dichotic listening task:
 - Information from both ears can pass through.
 - Unattended information is weakened but can **still receive further processing**.
- Words have different activation **thresholds**
 - Attenuated signals often fail to reach a word's threshold.
 - Relevant items, such as one's own name, have *lower thresholds* and may still activate.
(which explains the cocktail party effect)

Attenuation Model 衰减模型

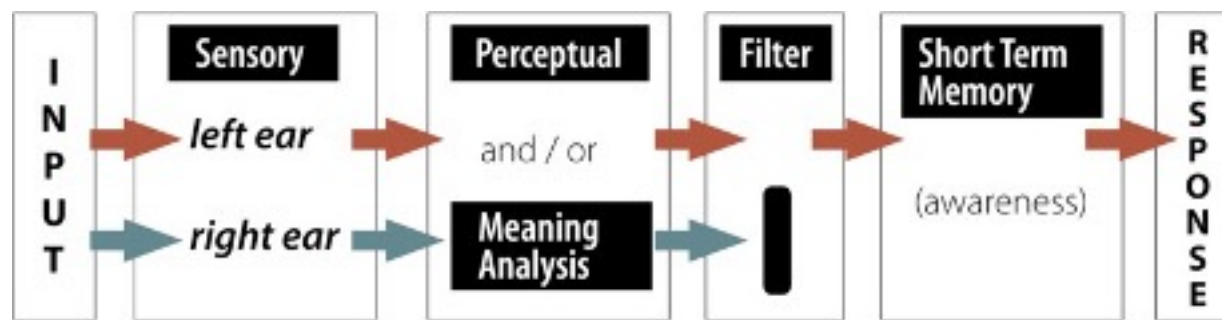
Treisman's Attenuation Model



- Attenuation replaces complete blocking. Information from unattended channels can still influence processing.
- Shared assumptions with the filter model:
 - 1) Higher-level analysis has limited capacity.
 - 2) Selection occurs between initial analysis and analysis of meaning.
 - 3) Both models regulate information at an **early** perceptual stage.

Late Selection Model

- **Deutsch and Deutsch's model** places selection after analysis of meaning, rather than during initial sensory processing.
- Both attended and unattended inputs can undergo semantic analysis.
All incoming sensory stimuli are fully processed for semantic meaning.
- Relevance then determines which information gains conscious access or guides a response.



[Image source](#)

Source: https://en.wikipedia.org/wiki/Broadbent%27s_filter_model_of_attention-Late_selection_models_of_attention

Evidence for late selection model

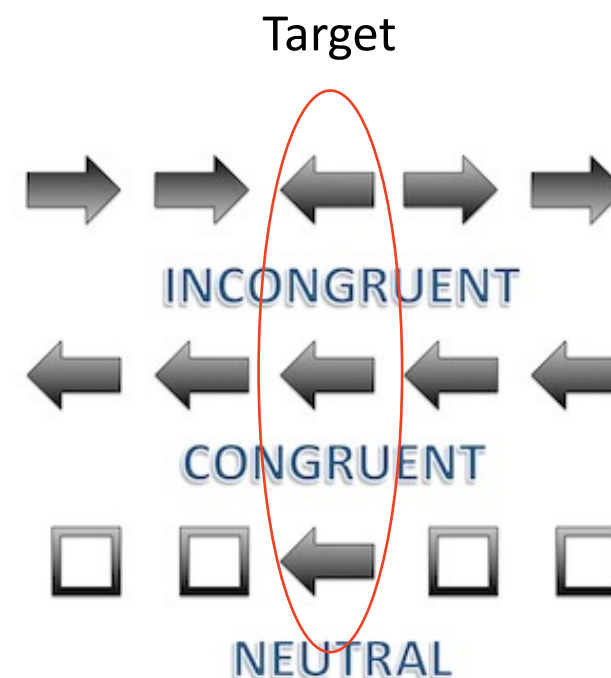
- In a dichotic listening task, participants heard words mixed with digits.
- Left ear: “Dear – 7 – Jane”. Right ear: “9 – Aunt – 6”
- According to the **early selection** model, participants should have reported all items presented to one ear first, and the items presented to the other ear second.
- However, the researchers found that participants reported hearing, **“Dear – Aunt – Jane”** and **“9 – 7 – 6”**
- The result suggests that selection can depend on meaning, which requires processing beyond basic low-level physical features.

Flanker Task

A paradigm for visual attention study

n. 干扰物 (侧翼刺激)

- A task for measuring visual selective attention.
 - Respond to a target while ignoring flankers (干扰物).
 - Congruent condition: Same response as the target.
 - Incongruent condition: A competing response.
 - Neutral: Neither response.
- Responses are slower on incongruent trials.



Source: https://en.wikipedia.org/wiki/Eriksen_flanker_task

ERP Study of Flanker Task

Source: <https://doi.org/10.1111/j.1469-8986.1996.tb00425.x>

- N200 pattern in EEG is found after onset of incongruent stimuli.

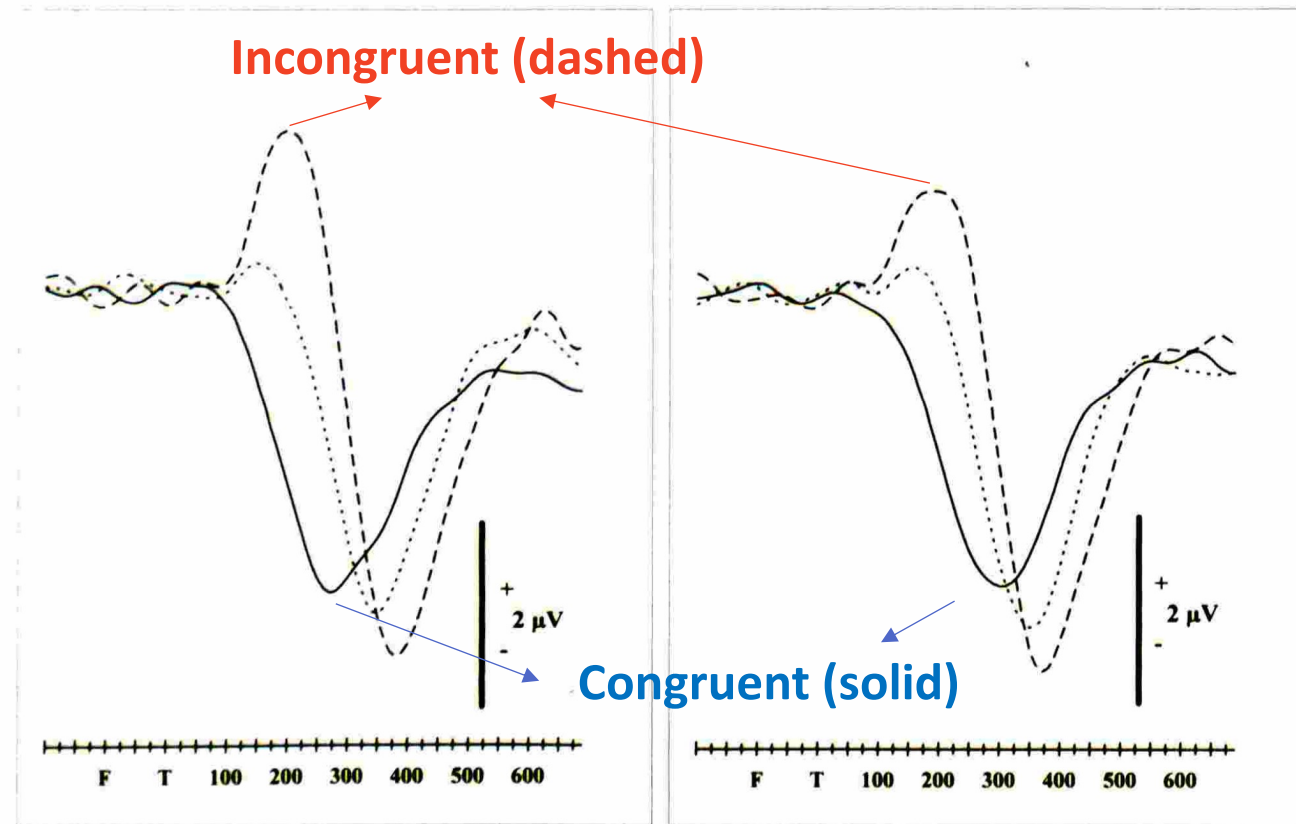
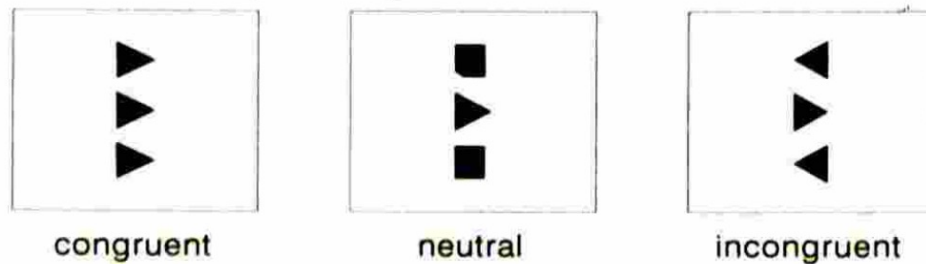
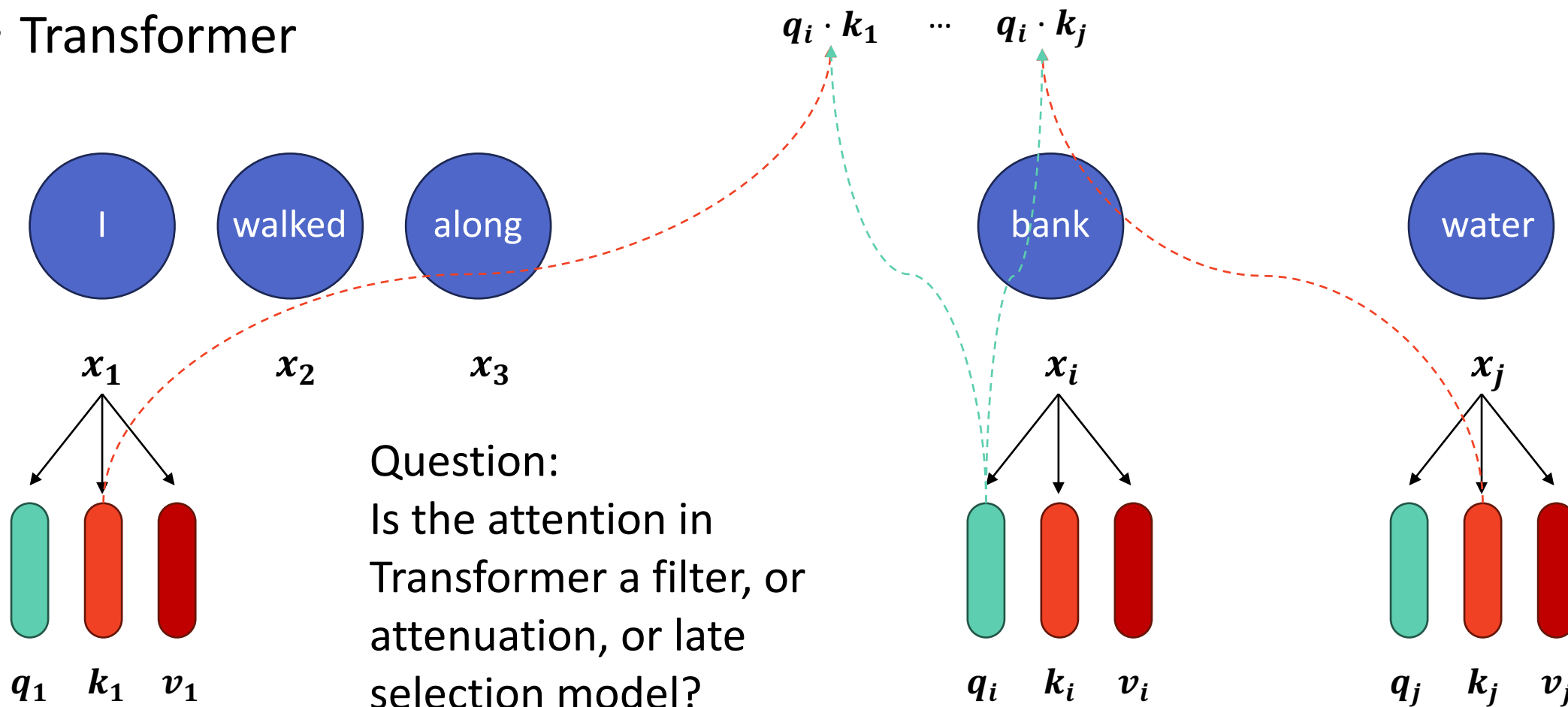


Figure 2. LRPs, averaged for flanker compatibility (congruent: solid line; neutral: dotted line; incongruent: dashed line) and spatial distance (short: left panel; large: right panel). F denotes the onset of flanker stimuli; T denotes the onset of target stimuli. The average waveforms were digitally filtered (low-pass cut-off frequency: 12 Hz).

Attention in AI

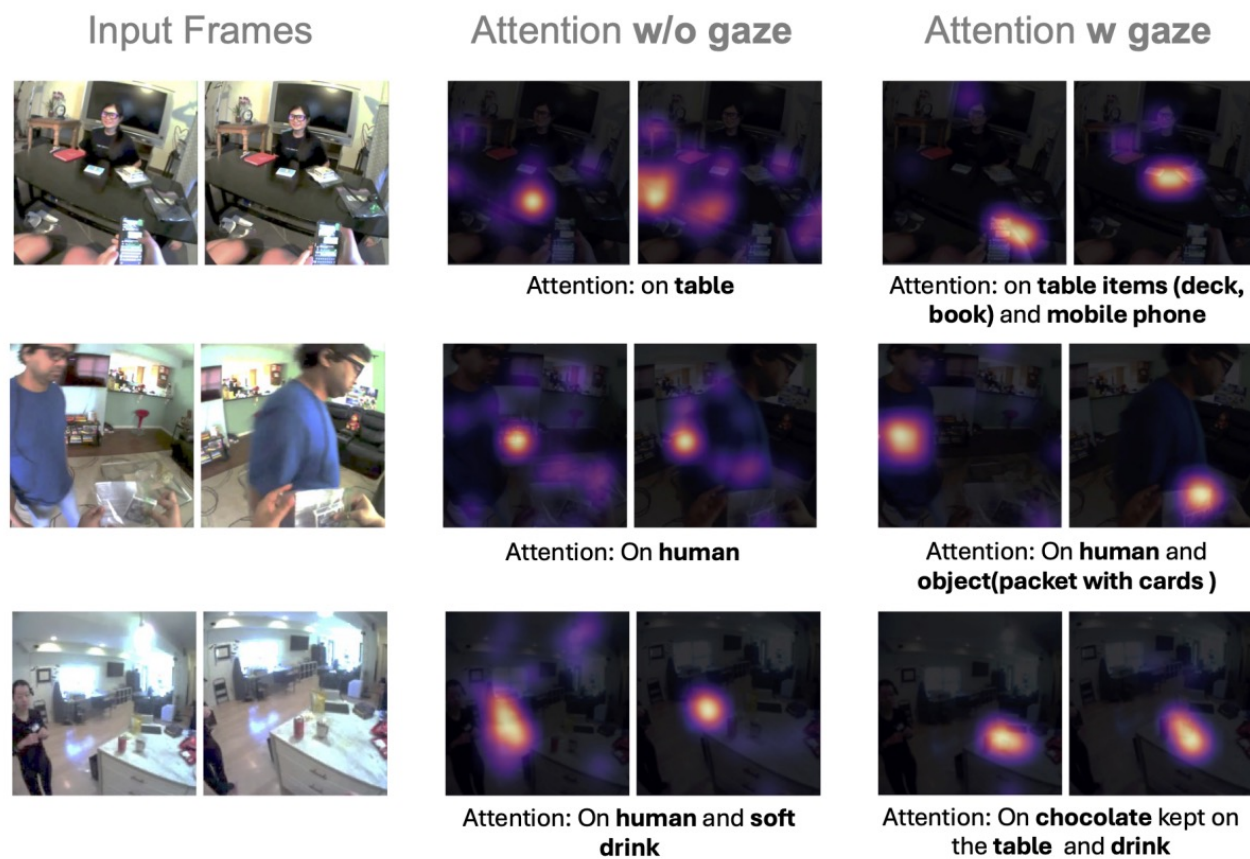
- Transformer

$$a_i = \sum_{j \leq i} \alpha_{ij} v_j$$



Gaze-VLM: Visual attention

Source: Pani & Yang (2025), Gaze-VLM · Figure 4



First-person video

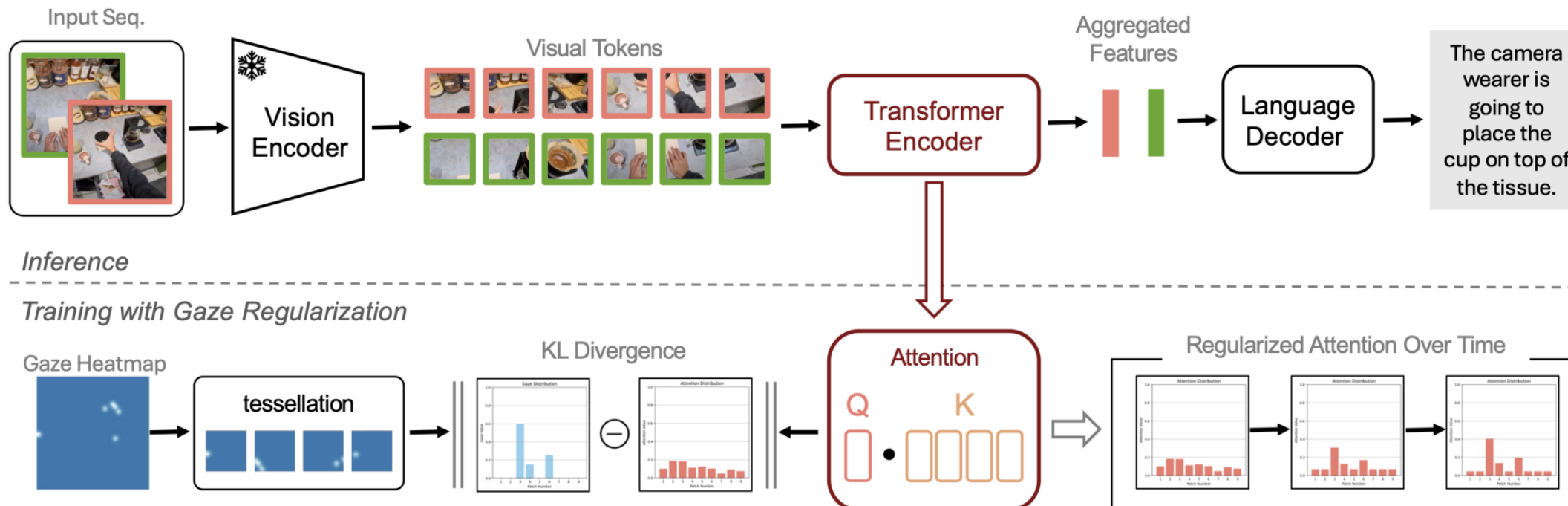
Describe ongoing activity.
Predict the next action.

Human gaze as a clue

Guide the model toward
objects relevant to the task.

Gaze-VLM: Guiding patch attention

Source: Pani & Yang (2025), Gaze-VLM · Figure 1



Training: match attention over image patches to the gaze heatmap.

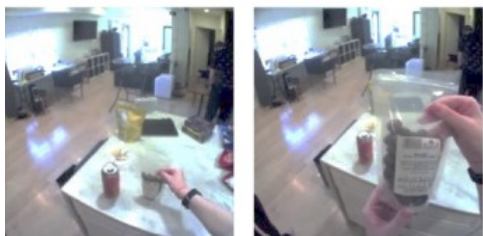
Use: describe the video without eye-tracking input.

Gaze-VLM: Better activity predictions

Source: Pani & Yang (2025), Gaze-VLM · Table 1 and Figure 3

OpenFlamingo on Ego4D · Semantic similarity (higher is better)

Task	Baseline	Gaze-guided
Future activity prediction	0.653	0.751
Current activity understanding	0.718	0.785



Future frames (reference)

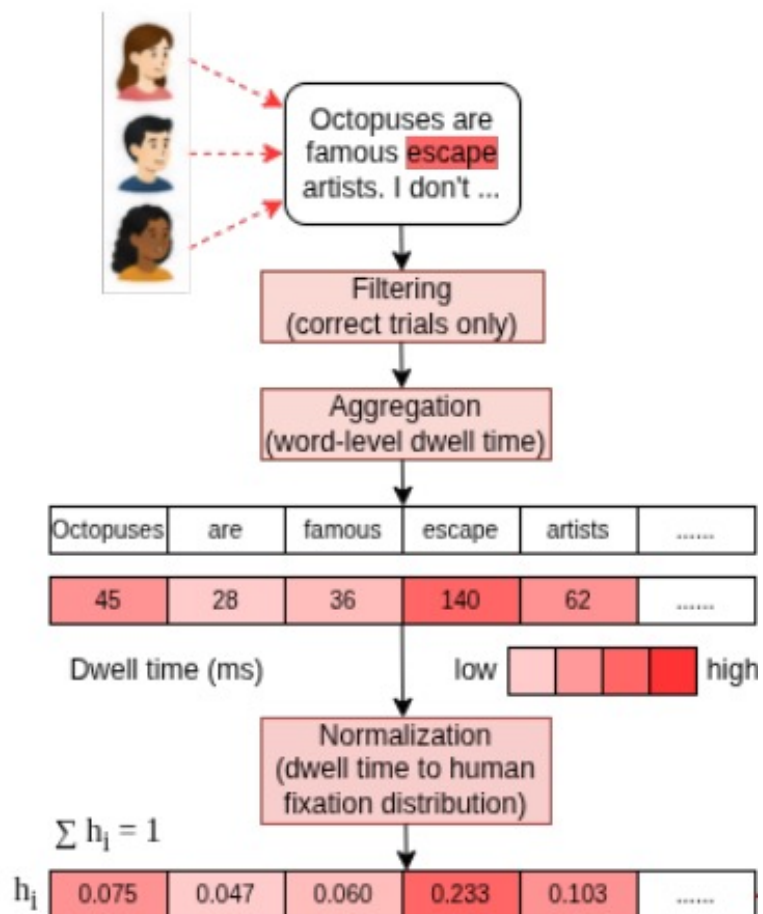
Baseline: reaching for a drink

Gaze-guided: reaching for chocolate

Gaze can reveal which objects matter for the person's next action.

Reading: Human fixation patterns

Source: Anonymous submission (2026), Aligning Decoder Attention with Human Fixations for Reading Comprehension · Figure 1 (detail)



OneStop reading data

Read the paragraph first.
See the question afterward.

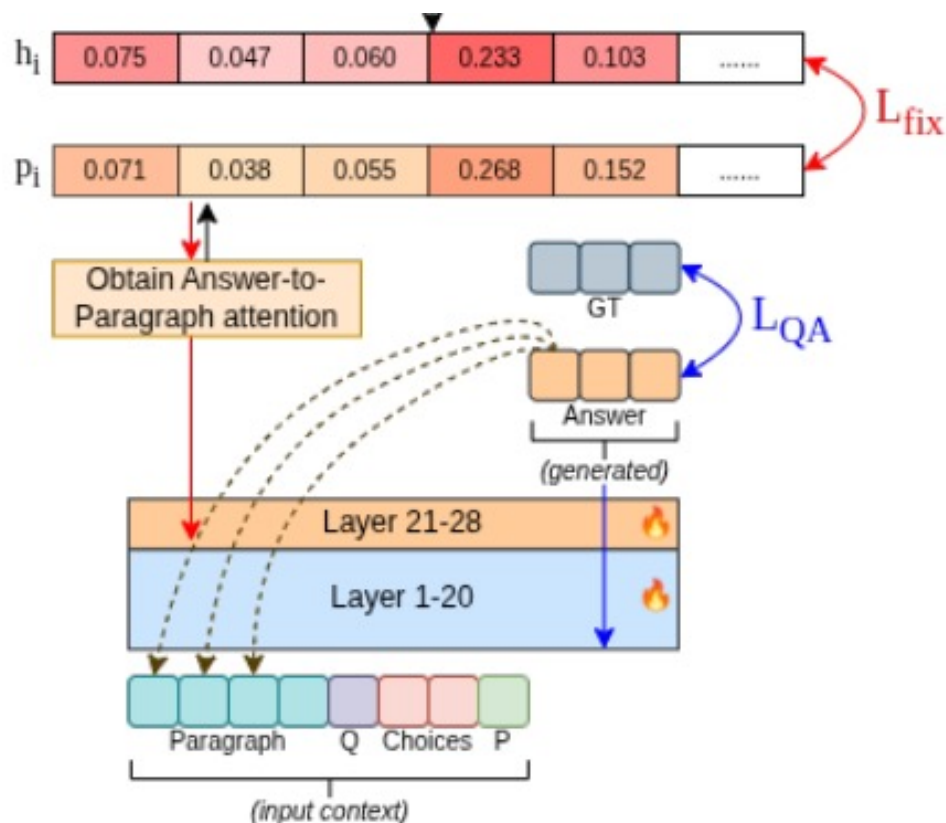
Build a word-level target

Average dwell time across readers
who answered correctly.
Normalize it across the words.

Longer fixation can also reflect difficulty.

Aligning decoder attention

Source: Anonymous submission (2026), Aligning Decoder Attention with Human Fixations for Reading Comprehension · Figure 1 (detail)



While generating an answer...

Collect attention to paragraph words from the final 8 decoder layers.

Two training goals

Answer the question correctly.
Match the human fixation distribution.

No eye tracking needed at test time.

Human fixations improve QA

Source: Anonymous submission (2026), Aligning Decoder Attention with Human Fixations for Reading Comprehension · Tables 3–4

Qwen3-0.6B · Three paragraph-disjoint splits · OneStop

Measure	QA only	QA + fixation
Four-choice QA accuracy	86.17%	89.45%
Human–model correlation (Pearson)	0.228	0.812

+3.28 percentage points in QA accuracy

Shuffling fixations weakens the benefit: word locations matter.

Human attention can guide model attention.

The evidence does not establish that their mechanisms are the same.